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✧ Contact

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✧ Personal Details

Given/Family Name: Chuang Yang | **Gender**: Male | **Birth**: 1992 | **Marital status**: Single
Current status: 3rd-year PhD Student at NUAA (5-year full-time doctor's degree program).
Language: Mandarin Chinese, English (professional proficiency).
Research interests:
Swarm perception, awareness and navigation in Robotics and Neuroscience; Neural basis of spatial cognition and navigation; Self-organizing behavior and complex dynamical processes in biological and nervous systems; Cooperative Navigation; Integrated Navigation System (INS/GNSS/Magnetic); Neuromorphic Swarm; Bio-inspired self-organizing and collective robots.
Journal Reviewer:
Chinese Journal of Aeronautics.

Professional association:

Member of Royal Institute of Navigation, Member of Chinese Society of Aeronautics and Astronautics, Member of Chinese Association for Artificial Intelligence, Member of Chinese Association of Automation.

✧ Current Research

- **Major**: Control Science and Engineering
- **PhD topic:** Brain-inspired swarm navigation for autonomous collective robots

  Abstract: This project (www.neuromorphicswarm.com) will fuse social perception/communication/ collective behavior of animal groups, neural basis of spatial representations of self and other in the hippocampus-entorhinal system, and the decentralized cooperative navigation in robotics by employing neural computing (e.g. CANN, ML, SNN) to develop a new class of brain-inspired swarm perception, awareness and navigation architecture——towards neuromorphic devices——in large-scale 3D environments for Neuromorphic-Swarm-Robot that navigate in 3D space as social animals do.

✧ Skill Profile

- **Programming**

  MATLAB, C, LabVIEW, Python
- **Web Design**

  Hugo, Github
- **Open Source Program**

  RatSLAM: a brain-based SLAM system

  NeuroSLAM: a brain-inspired SLAM system for 3D environments

  SwarmLab: an open source MATLAB Drone Swarm Simulator

  GNSS/INS: an Integrated Navigation System Simulator (designed by my group).
- **Embedded Hardware**

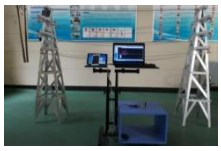
  Arduino, STM32
- **Design of Electronic System**

  (***I became interested in electronic products in childhood, greatly influenced by my parents who were all maintanance workers for electronic devices, such as television, amplifier, etc.***)

  Two seclected designs of electronic system:

IMU/Magnetic Sensors-based Integrated Navigation System of Measurement While Drilling

Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate (***team leader***), 09/2017.



IMU-based On-line Monitoring System of Transmission Line Galloping

Second Prize in the 12th Electronic Design Competition for China postgraduate (***team leader***), 07/2017.

✧ Education Experiences

- **04/2018—present**

  Nanjing University of Aeronautics and Astronautics, China

  PhD student of Control Science and Engineering

  Supervisor: Prof. Jianye Liu (***director of Navigation Research Center of NUAA***)

  Co-supervisor: Prof. Zhi Xiong

  Thesis Topic: *Brain-inspired swarm navigation for autonomous collective robots*

- **07/2020—07/2020**
  Certificate of Completion of Neuromatch Academy (an online school for Computational Neuroscience)
  Interactive student of Computational Neuroscience
  Mentor: Dr. Torres Gelsy (Universtity of Pittsburgh)
  Course Project: *How does task related activity interact with intrinsic activity of the brain*?
- **09/2015—12/2017**
  Henan Polytechnic University, China
  Master of Control Science and Engineering (***graduating half a year in advance***)
  Supervisor: Dr. Jinxian Yang
  Final thesis: *Research on INS/magnetic integrated navigation technique for measurement while drilling*
- **09/2011—07/2015**
  Henan Institute of Science and Technology, China
  Bachelor of Electrical Engineering and Automation (GPA: 3.68/4.0)

✧ Seclected Project Experiences

- **Brain-inspired navigation & swarm navigation**

1. China, National Natural Science Foundation of China, Grant No. 61873125. **Brain-inspired intelligent cooperative navigation for autonomous collective robots,** 01/2019-12/2022. (main participant, my PhD topic).

- **Integrated navigation**

2. China, National Natural Science Foundation of China, Grant No. U1404510. **Key technologies of MEMS inertial navigation for measurement while drilling**, 01/2015-12/2017. (main participant)
3. China, Key Science and Technology Program of Henan Province, Grant No. 172102210289. **Research on on-line monitoring system of transmission line galloping based on inertial technology**, 09/2016-09/2017. (main participant)

✧ Publications

- **Brain-inspired navigation & swarm navigation**

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, Lijun Chao, Yudi Chen. **A 3D path integration approach based on multi-scale grid cells for large-scale navigation**[J]. IEEE Transactions on Cognitive and Developmental Systems (in revision, 2020.11.24).
2. **Chuang Yang**, Jianye Liu, Zhi Xiong, Lijun Chao. **Brain-inspired vector navigation method based on model of multi-scale grid cells**[J]. Journal of Chinese Inertial Technology, 2020, 28(02):179-185. (EI, in Chinese)
3. **Chuang Yang**, Jianye Liu, Zhi Xiong. **A new paradigm of brain-inspired navigation integrating perception and decision**[EB/OL]. article.xuexi.cn, 2020.03.04. (in Chinese)
4. **Chuang Yang**, Jianye Liu, Zhi Xiong, Jun Xiong. **Brain-inspired navigation technology integrating perception and action decision: A review and outlook**[J]. Acta Aeronautica et Astronautica Sinica, 2020, 41(1):323280. doi:10.7527/S1000-6893.2019.23280. (EI, in Chinese)
5. Jianye Liu, **Chuang Yang**, Zhi Xiong, Jizhou Lai, Jun Xiong. **Attractor neural network-based brain-inspired navigation technology**[J]. Navigation Positioning & Timing, 2019, 6(5): 52-60. (in Chinese)
6. **Chuang Yang**, Jianye Liu. **Brain-inspired navigation technology and its applications in unmanned autonomous system**[J]. Naas & Inertal Technology, 2019, 200+201:151-152. (in Chinese)

- **Integrated navigation**

7. **Chuang Yang**, Qinghua Zeng, Jinxian Yang, Jinbao Jiang, Andrew H. Kemp. A gyro-aided rotary norm constraint filtering method correcting geomagnetic azimuth while drilling[J]. IEEE Sensors Journal. (to

be submitted, 2020.11)

8. Jinxian Yang, **Chuang Yang**, Shuanglei Li, et al. **Vibration analysis of transmission lines based on MIMU**[J]. Journal of Vibration and Shock, 2018, 37(11): 230-236. (EI, in Chinese)
9. Jinxian Yang, **Chuang Yang**, Jinbao Jiang. **A robust mag/INS-based orientation estimation algorithm for measurement while drilling**[J]. IEEE Sensors Journal, 2017, 17(23): 7971–7980. (SCI)
10. Jinxian Yang, **Chuang Yang**. **MEMS gyro signal reconstruction via layerwise and adaptive MP**[J]. Optics and Precision Engineering, 2017, 25(12): 3160-3168. (EI, in Chinese)
11. Jinxian Yang, **Chuang Yang**, Zhitao Jiang. **Adaptive extraction of gravity information for drilling tool with magnetic and inertial integrated system**[J]. Journal of china coal society, 2017, 42(12): 3331-3337. (EI, in Chinese)
12. Jinxian Yang, **Chuang Yang**. **Signal extraction of mems gyro utilized in MWD based on sparse representation**[J]. Chinese Journal of Sensors & Actuators, 2017, 30(11): 1677-1683. (in Chinese)
13. Jinxian Yang, **Chuang Yang**. **Gyro drift estimation and compensation based on gravity quaternion**[J]. Chinese Journal of Sensors and Actuators, 2017, 30(8): 1187-1192. (in Chinese)

✧ Patents

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, et al. **A brain-inspired navigation method in 3D environment based on path integration of multi scale grid cells**[P]. China Invention Patent, (2020-08, Under review)

✧ Honors and Awards

10/2017: National Scholarship for Postgraduate

09/2017: Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate, ***INS/Magnetic-based integrated navigation system of measurement while drilling***. (***team leader***)

07/2017: Second Prize in the 12th Electronic Design Competition for China postgraduate, ***INS-based on-line monitoring system of transmission line galloping***. (***team leader***)

09/2015-01/2018: Full scholarship for Postgraduate

11/2013: National Scholarship for Encouragement

03/2013: Third-level Certificate for National Computer

11/2012: National Scholarship for Encouragement

09/2011-07/2015: The First Prize Scholarship for four years